Joint Detection of AI-Generated Images and Post-Processing Alterations in Real-World Scenarios

Bridging the gap between ideal forensic detection and real–world robustness

RRDataset

ViT / ResNet

Two Heads

Cross-Class Analysis

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Bridging the gap between ideal forensic detection and real-world robustness.

PRESENTATION ROADMAP:

- Problem and motivation
- Related work and gap
- Proposed shared-backbone + two-head method
- RRDataset and preprocessing
- Experimental setup and metrics
- Results, ablations, and confusion matrices
- Cross-class transformation analysis
- Conclusions, future work, and references

Core message

-The project tests whether one shared visual representation can jointly answer two questions:

-Is the image real or AI-generated?
-Which post-processing path produced the observed image?

Problem Statement : The Generalization Gap

The Core Problem:

AI-generated images are increasingly circulated through realistic pipelines: social platforms, compression, screenshots, printing, scanning and many more. Current detectors achieve saturated performance on "clean" data but fail in practical deployment.

Primary Challenges:

- Transmission: Multiple compression cycles (WhatsApp, Telegram, X).
- Re-digitization: Physical media tampering (Print-Scan, Screen-Shooting).

Limitations of Standard Benchmarks:

- Artificial distortions (e.g., Gaussian blur) do not replicate authentic degradation.
- Bias toward everyday-life imagery ignores specialized, high-risk scenarios.

Why it matters:

–Better evidence quality for journalism, public safety, and social–media verification.

–More realistic evaluation of robustness against image distribution channels.

–Understanding transformation effects can reveal detector failure modes.

The "Trust Crisis" Statistics:

- 89.31% of observers in specialized scenarios automatically categorize genuine images as AI-generated when faced with unfamiliar artifacts.

Related Work & State of the Art

RRDataset (Li et al., ICCV 2025)

The first benchmark to address re-digitization and multi-cycle internet transmission.

HAMMER Framework (Shao et al., CVPR 2023)

Proved that multi-task grounding—identifying how a manipulation occurred—improves macro classification accuracy.

Architectural Evolution & Collapse

Baselines (ResNet/DIRE) achieve high accuracy on clean data but suffer total collapse under re-digitization (DIRE fake accuracy drops from 89.72% to 1.42%).

Proposed Shift: Transitioning from low-level pixel features to semantic multi-scale aggregation.

Shared Backbone & Dual-Head Architecture

Input Image 224x224 RGB -> ViT-small_patch16_224 (Global Self-Attention)

-> Head 1: fc_rf (Real / AI) -> Binary source authentication.

-> Head 2: fc_tr (Transformation) -> Multi-class tracking (Original, Transmitted, Re-digitized).

Strategic Rationale: Convolutional kernels are architecturally predisposed to local texture bias, making them vulnerable to frequency loss in transmission. ViT self-attention captures global inconsistencies across texture and spatial regions.

Kendall Uncertainty Weighting

$$L_{total} = \frac{1}{2\sigma_1^2} L_{rf} + \frac{1}{2\sigma_2^2} L_{tr} + \log \sigma_1 + \log \sigma_2 >$$

The Weighting Dilemma:

Manual tuning of l_{rf} and l_{tr} is suboptimal and unstable due to the inherent noise variance in re-digitized samples.

The Solution:

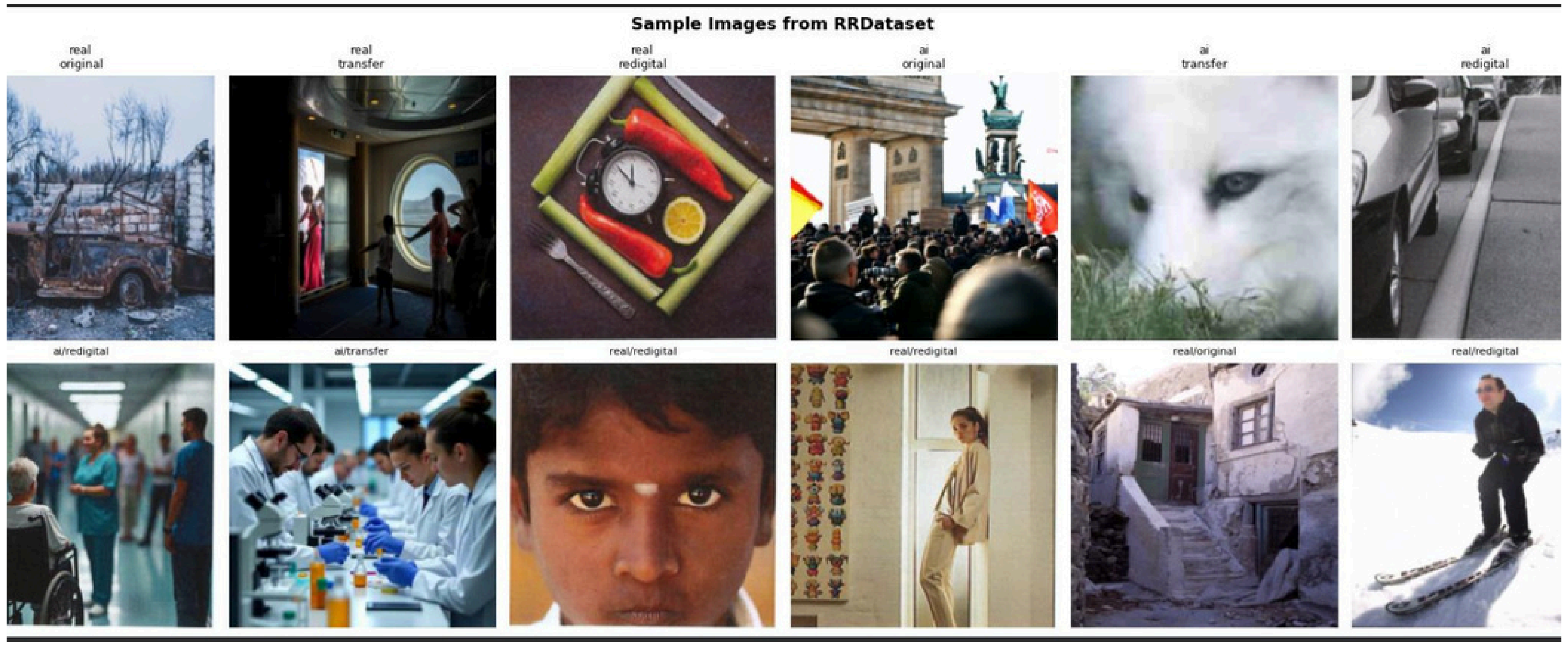
Homoscedastic uncertainty implements learnable log-variance parameters. The model automatically prioritizes "easier" tasks during early training to stabilize gradients, allowing the transformation head to assist the real/fake head without overwhelming the process.

Dataset: The RRDataset Benchmark

SCENARIO	STRUCTURAL AXIS (TRANSFORMATIONS)
War & Conflict / Medical	Original: Clean, source data
Political / Culture	Transmitted: Social Media (WhatsApp, Facebook, X)
Labor / Disasters	Re-digitized: Print-Scan, Screen-Shooting

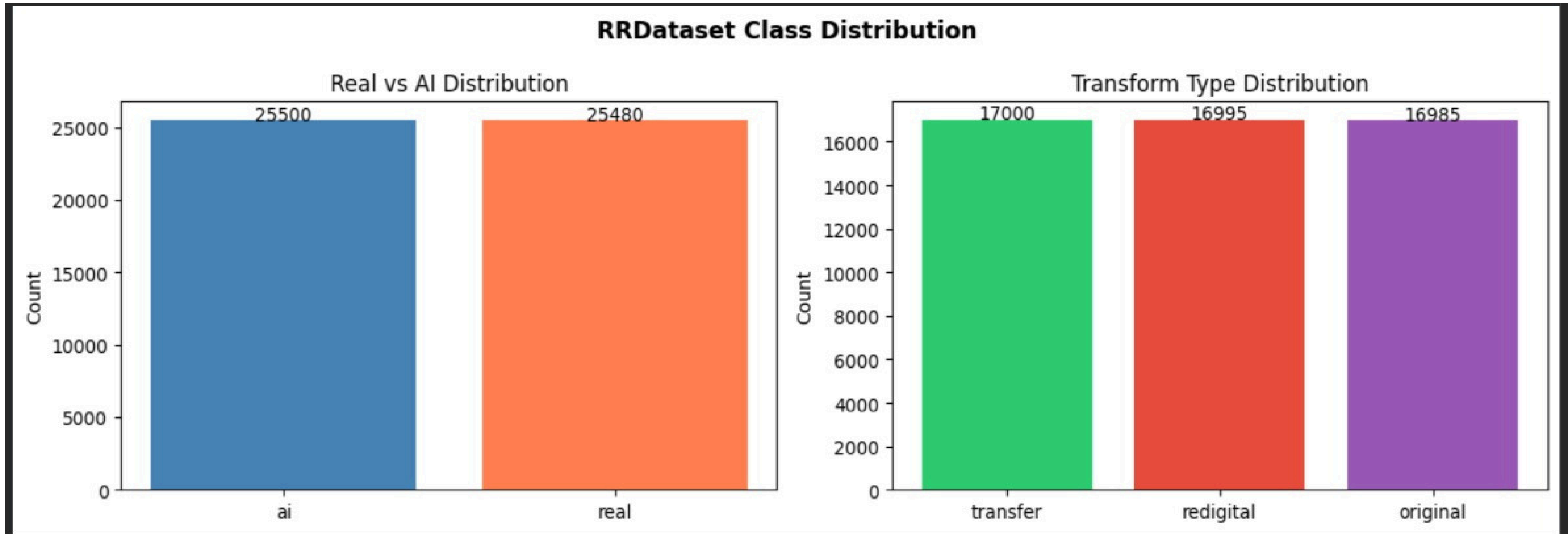
Scale: Total RRDataset contains ~50,000 images.
Subset Selection: 20,000 balanced images (10,000 Real / 10,000 AI) were used for training to ensure class-invariant feature learning and prevent "suspicion bias."

Dataset



- Original: unmodified benchmark images.
- Transfer: images after internet/social platform transmission.
- Redigital: images after physical or screen-based recapture.
- Image transforms: resize to 224, augmentation for training,
- ImageNET normalization

- Classes: real vs AI-generated.
- Transformation labels: original, transfer, redigital.
- Inventory from notebook: 50,980 images.
- Split: 35,686 train / 7,647 validation / 7,647 test.
- Near-balanced across real/fake and transformation classes.



Experimental Setup

- Configuration: ViT_small_patch16_224 (PyTorch).
- Training Parameters: 224×224 input; RF-only, TR-only, equal, RF-heavy, TR-heavy, and learned uncertainty weighting were compared.
- Evaluation Metrics:
  - Accuracy: Overall classification success.
  - F1-Score: Critical for judging classification bias in the context of the "Trust Crisis."
- Baseline: ResNet50 trained on unimodal tasks.

Compute Note: Experiments were run on a debug-sized subset with limited epochs due to GPU constraints.

Results: Backbone Performance Comparison

Model / Run	RF Acc	RF F1	TR Acc	TR F1
ResNet50 RF	0.6200	0.5786	-	-
ViT RF	0.8200	0.8193	-	-
Equal multitask	0.8200	0.8199	0.5000	0.4994
RF-heavy final	0.7800 val / 0.8367 test	0.7792 val / 0.8357 test	0.4400 val / 0.4833 test	0.4297 val / 0.4789 test
Uncertainty	0.8067	0.8065	0.5067	0.5081

Baseline Results

```python
# show baselines
pd.read_csv('results/baselines.csv')
```

	backbone	task	rf_acc	rf_f1	tr_acc	tr_f1
0	resnet50	realfake	0.6200	0.5786	0.0000	0.0000
1	resnet50	transform	0.5200	0.5113	0.4067	0.3883
2	vit_small_patch16_224	realfake	0.8200	0.8193	0.0000	0.0000
3	vit_small_patch16_224	transform	0.5067	0.4504	0.4800	0.4676

- Best baseline in the shown runs: ViT real/fake head reaches 0.8200 accuracy and 0.8193 macro F1.
- ResNet50 real/fake baseline reaches 0.6200 accuracy.
- Transformation-only runs are harder, around 0.48-0.52 accuracy in the shown debug/validation outputs.

```
debug: baseline runner ready

>>> resnet50_rf | backbone=resnet50 | mt=False
ep 1/3 | loss=0.6812 | rf_acc=0.5267
ep 2/3 | loss=0.6442 | rf_acc=0.6067
ep 3/3 | loss=0.6227 | rf_acc=0.6200

>>> resnet50_tr | backbone=resnet50 | mt=True
ep 1/3 | loss=1.0941 | rf_acc=0.4600 | tr_acc=0.3733
ep 2/3 | loss=1.0762 | rf_acc=0.4933 | tr_acc=0.3600
ep 3/3 | loss=1.0620 | rf_acc=0.5200 | tr_acc=0.4067

>>> vit_rf | backbone=vit_small_patch16_224 | mt=False
Unexpected keys (norm.bias, norm.weight) found while loading pretrained weights. This may be expected if model is being adapted.
ep 1/3 | loss=0.4649 | rf_acc=0.7200
ep 2/3 | loss=0.1901 | rf_acc=0.7800
ep 3/3 | loss=0.0610 | rf_acc=0.8200

>>> vit_tr | backbone=vit_small_patch16_224 | mt=True
Unexpected keys (norm.bias, norm.weight) found while loading pretrained weights. This may be expected if model is being adapted.
ep 1/3 | loss=1.1150 | rf_acc=0.5067 | tr_acc=0.4800
ep 2/3 | loss=0.8209 | rf_acc=0.4800 | tr_acc=0.5200
ep 3/3 | loss=0.5486 | rf_acc=0.4933 | tr_acc=0.5800
```

Multi –Task Ablation

```
# show multitask
pd.read_csv('results/multitask.csv')
```

	run	backbone	w_rf	w_tr	rf_acc	rf_f1	tr_acc	tr_f1
0	multitask_A_rf_only	vit_small_patch16_224	1.0	0.0	0.8200	0.8193	0.3400	0.2820
1	multitask_B_tr_only	vit_small_patch16_224	0.0	1.0	0.5067	0.4504	0.4800	0.4676
2	multitask_C_equal	vit_small_patch16_224	0.5	0.5	0.8200	0.8199	0.5000	0.4994
3	multitask_D_rf_heavy	vit_small_patch16_224	0.7	0.3	0.7800	0.7792	0.4400	0.4297
4	multitask_E_tr_heavy	vit_small_patch16_224	0.3	0.7	0.7600	0.7596	0.4933	0.4682
5	multitask_F_uncertainty	vit_small_patch16_224	learned	learned	0.8067	0.8065	0.5067	0.5081

Best RF acc shown
0. 820 0

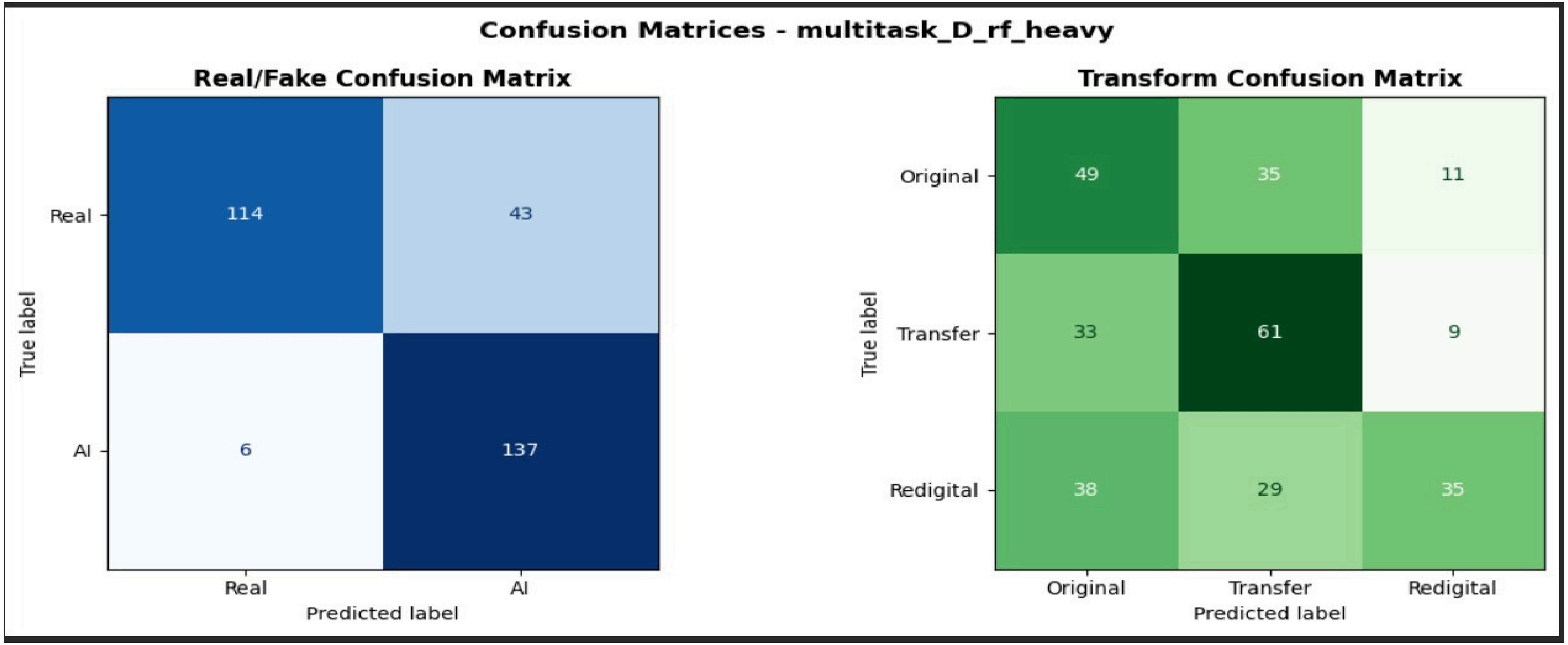
Best transform acc shown 0. 506 7

Selected final run
RF-heavy

- **Run 2 (Equal Weighting):** *Peak RF F1 (0.8199) but less stable joint performance*. Equal weighting: strong real/fake performance with higher transform accuracy than task-isolated real/fake training.
- **Run 4 (RF Heavy):** *RF Acc: 0.7800 / TR Acc: 0.4400*. RF-heavy weighting selected as final evaluated run in the notebook: multitask_D_rf_heavy.
- **Run 5 (Uncertainty):** *RF Acc: 0.8067 / TR Acc: 0.5067.* Uncertainty weighting balances the tasks automatically and reaches the highest transform accuracy shown.

Note: We tested both fixed weighting and learned uncertainty weighting. RF-heavy was selected for the final confusion-matrix evaluation, while uncertainty weighting achieved the best validation transformation accuracy.

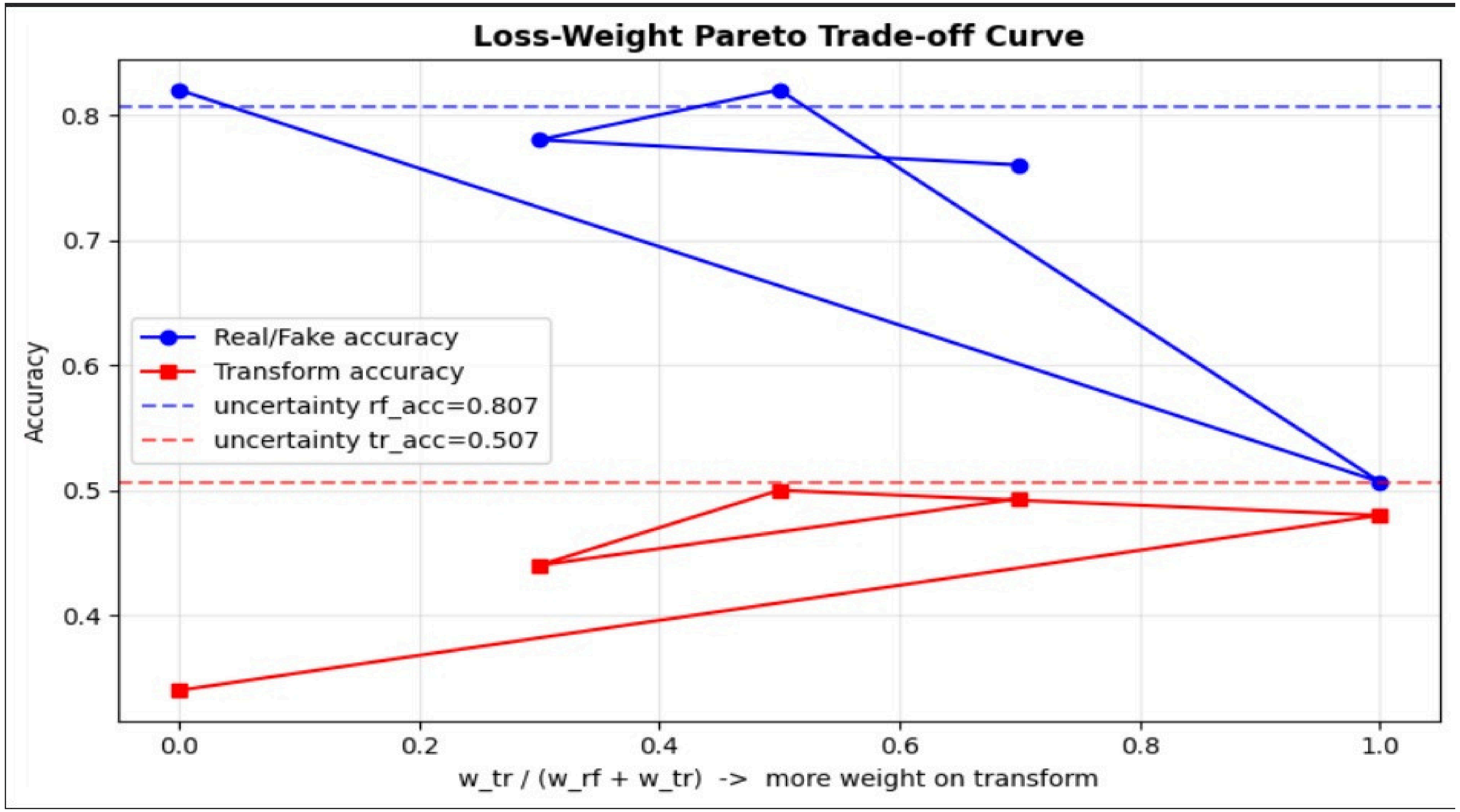
Confusion Matrices & Error Analysis



Real/fake detection is much cleaner than transformation recognition; the transform head most often confuses visually related post-processing paths.

Note: Results are preliminary pipeline results, not full-scale RRDataset benchmark numbers.

Loss-Weight Trade-Off



Takeaway: report both heads, not one aggregate score

- **AI Detection Performance:** Excellent recall (137/143 AI samples correctly identified).
- **The "Suspicion" Bias:** The model is more "suspicious" of real images, correctly identifying only 114/157 Real samples. This mirrors the human "Trust Crisis."
- **Transformation Bottleneck:** Primary confusion persists between "Original" and "Transfer" labels. Re-digitization remains the most damaging operation for classification consistency.

Cross-Class Transformation Analysis

```
•••  realfake        ai     real
     transform
     original    0.9767  0.6154
     redigital   0.9273  0.8298
     transfer    0.9778  0.7414
     original: real=0.615, ai=0.977, delta=+0.361
     transfer: real=0.741, ai=0.978, delta=+0.236
     redigital: real=0.830, ai=0.927, delta=+0.098
```

- The final cross-check compares real/fake accuracy inside each transformation type.
- AI images are easier for the model across all listed transformations.
- The largest gap appears for original images: AI accuracy 0.977 vs real accuracy 0.615.
- The smallest gap appears for redigital images: AI accuracy 0.927 vs real accuracy 0.829

Original
Delta +0.361

Transfer
Delta +0.236

Redigital
Delta +0.098

Interpretation: transformation category changes the error profile, so robustness should be evaluated per transformation rather than only with a single overall score.

Conclusions and Future Work

1. **Backbone Dominance:** ViT's self-attention mechanism is a prerequisite for robust forensic detection in degraded environments.
2. **Optimization Stability:** Learned uncertainty weighting is the only effective method for balancing the disparate noise levels of joint tasks.
3. **The Re-digitization Wall:** Physical conversion remains the single most significant challenge for forensic generalization.
4. **Addressing the Trust Crisis:** We have developed a framework that acknowledges and partially mitigates the tendency to misclassify genuine images.

1. **VLM Hybridization:** Integrating ViT forensic heads with Vision-Language Model (GPT-4o) semantic reasoning to combat the 27.07% re-digitization drop seen in zero-shot VLMs.
2. **Frequency-Domain Fusion:** Implementing spectral analysis to "see through" re-digitization artifacts that cause local feature collapse.
3. **Platform-Specific Tracking:** Expanding the transformation head to detect specific social media footprints (e.g., WhatsApp vs. Instagram).
4. **Physical Media Loop:** Finalizing the robustness loop by investigating physical media-tampering signatures (e.g., print-scan patterns).

References:

- Li, C., et al. (2025). "Bridging the Gap Between Ideal and Real-world Evaluation." ICCV.
- Shao, R., et al. (2023). "Detecting and Grounding Multi-Modal Media Manipulation." CVPR.
- Kendall, A., et al. (2018). "Multi-Task Learning Using Uncertainty to Weigh Losses." CVPR.

THANK YOU!